

s1.04 — Health monitoring for millions of edge devices

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Interview framing

- Target: OCI IC3 ownership of a bounded platform component.
 - `IC3 baseline` is what must land in every phase.
 - `IC4 stretch` adds architecture changes under stronger constraints.
 - These are preparation labels, not Oracle's official rubric.
- Evidence: retain the first-hand report attribution in [the bank](#).
 - The architecture below is a proposed interview answer.
 - It does not claim to reproduce Oracle's implementation or an interviewer's accepted answer.
- Why this is hard
 - Naive: a `last_seen` column per device, a cron scan for stale rows, and a ping to confirm.
 - First break: 1 million devices at 15 s is 66,667 writes/s plus a full scan every few seconds, and devices behind NAT cannot be pinged.
 - Second break: a timer fires while a fresh heartbeat lands, or one collector dies and 100,000 devices go silent at once
 - Stale expiry overwrites fresh evidence and monitoring failure looks like device failure.
 - Minimum primitives
 - A registry of expected devices
 - Heartbeats ordered by incarnation and sequence
 - A per-shard deadline index with version-checked expiry
 - Ownership epochs
 - A separate coverage watchdog that can say `UNKNOWN`.
- 🗨️ "Silence is missing evidence, not proof; the design makes suspicion versioned and monitoring blindness visible."
- Opening script
 - "I will treat this as a registry-driven heartbeat system with a versioned deadline index and an explicit distinction between device suspicion and monitoring failure."
 - "First I confirm fleet size, heartbeat cadence, NAT constraints, and how fast a silent device must be reported."
 - "Then I size ingest, hot state, and history, which decides sharding and why transitions rather than raw heartbeats are persisted."

- “The deep dive is the deadline algorithm: how a stale timer or an old collector owner is prevented from overwriting fresh evidence.”
- “I will close with what happens when the monitoring itself fails, alert grouping, and where the design changes at ten times the fleet.”

Phase	Clock	Deliverable
Intro and behavioral slice	00:00–05:00	90-second pitch; confirm the prompt
Requirements	05:00–11:00	Scope, clarifying answers, primary invariant
Estimates and contract	11:00–17:00	Heartbeat load and hot state; API semantics
Architecture	17:00–30:00	Diagram, shard ownership, one heartbeat trace
Deep dive	30:00–45:00	Deadline index, version fencing, owner epochs
Operations and tradeoffs	45:00–50:00	Failure table, coverage watchdog, tradeoff, wrap script
Wrap-up	50:00–54:00	Four-sentence recap and weakest assumption
Questions for interviewer	54:00–60:00	Two role-specific questions

- Clock note
 - If the interviewer states 45 minutes, drop the intro to 2 minutes and compress each phase proportionally.
 - The interviewer's clock wins over this table.

Requirements — IC3 baseline

- Numerical targets are interview assumptions unless an adjacent primary source states otherwise.

Question	Assumed answer	Why it changes the design
Fleet size and cadence?	1 million devices, 15 s heartbeat with jitter	Fixes ingest rate, shard count, and history volume
Can the service reach devices?	Many sit behind NAT; only an outbound channel exists	Push is primary; pull is a bounded secondary check
What does "health" mean?	Liveness of an authenticated agent; metrics live elsewhere	Keeps payloads small and this service separate from ingestion
Detection target?	Suspect after 45 s, published within 60 s	Sets timeout, tick size, and the expiry-work budget
What happens on detection?	Alert and dashboard only; no automated remediation	Removes the fencing burden from this service; keeps it a non-goal
How is a reboot identified?	Enrolled incarnation plus sequence per incarnation	Ordering cannot rely on device clocks or random boot IDs
Multi-tenant?	Yes, devices belong to tenants and sites	Scopes every read, quota, and alert group

- Functional scope
 - Enroll a device and its active incarnation.
 - Accept authenticated heartbeats and keep the latest observation.
 - Project a state per device: UP, SUSPECT, UNREACHABLE, UNKNOWN.
 - Suppress paging during maintenance without stopping observation.
 - List devices by state per tenant
 - Expose transition history.
 - Group alerts by shared collector, network, or region failure.
- Health is an observation, not certainty about physical failure.
 - Heartbeat presence indicates recent communication from an authenticated agent.
 - Silence may mean agent failure, network failure, or monitoring failure.
 - A failed verification probe does not prove that a device is dead.
- Assumed scope
 - 1 million registered devices.

- Some devices sit behind NAT and cannot receive unsolicited probes.
- Device liveness is primary.
- Detailed metrics use the separate ingestion service.

Requirement	Target	Enforced by
Heartbeat interval	15 s with jitter	Agent schedule; server rejects cadence below 5 s per device
Suspect threshold	45 s of silence	Deadline index per shard; version-checked expiry
Suspect publication	Within 60 s under healthy collector scheduling	5 s tick bound plus outbox drain budget
Grouped operator alert	Within 120 s when observation remains unavailable	Coverage watchdog independent of collectors
Heartbeat acceptance	99.95% regional	Stateless ingress fleet; shard failover with epochs
Ingest capacity	66,667 heartbeats/s sustained with 2× headroom	Ten logical shards plus standby instances
Dashboard read	p99 < 500 ms for a bounded fleet page	Cursor-paged state index; no fleet-wide scans
Stale expiry accepted	0 by construction	Conditional version compare under shard serialization

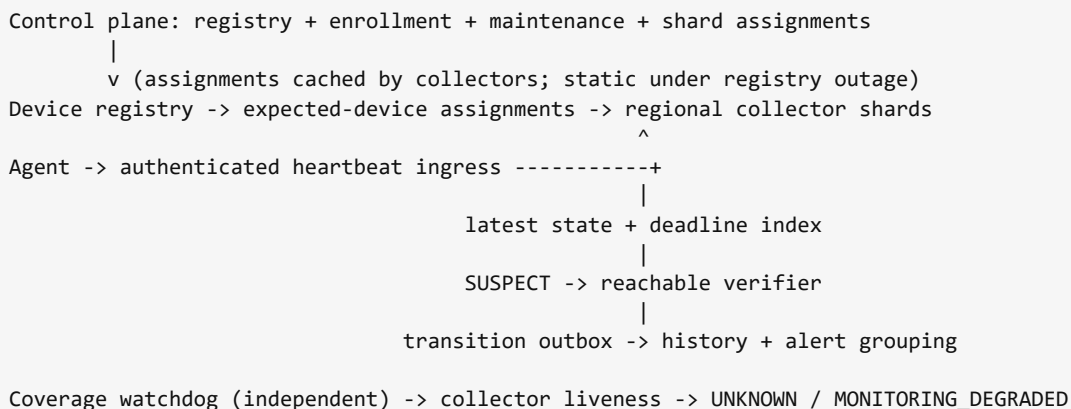
- Invariants
 - Registry defines which devices are expected to report.
 - An old heartbeat cannot overwrite a newer device incarnation/sequence.
 - A stale expiry task cannot mark a recently refreshed device suspect.
 - Loss of collector coverage must be visible as UNKNOWN or MONITORING_DEGRADED.
 - A transition is recorded before its projection is replaced, so evidence for an alert is never lost.
- Non-goals
 - Metrics and log ingestion.
 - Automated remediation or failover of the device's workload.
 - Proving physical failure from silence.
 - Firmware or configuration distribution.

Capacity and consequence — IC3 baseline

- Heartbeat load
 - $1,000,000 / 15 \approx 66,667$ heartbeats/s.
 - At 200 B payload, approximately 13.3 MB/s before transport overhead.
 - Ten logical shards each own about 100,000 devices and 6,667 heartbeats/s.
 - Extra instances supply failover and skew headroom.
 - Decision: shard by stable device identity with one active owner per shard
 - A single database row update per heartbeat would need 66,667 writes/s and is rejected.
- Hot state
 - At an assumed 128 B/device, logical state is about 128 MB fleet-wide.
 - Real indexes and runtime objects increase memory.
 - Deadline bookkeeping is manageable if old timer entries do not grow without bound.
 - Decision: keep latest state and the deadline index in memory per shard owner with a replicated checkpoint
 - The store is for recovery, not the hot path.
- Deadline work
 - 6,667 renewals/s per shard against a 100,000-entry ordered index is about $6,667 \times \log_2(100,000) \approx 113,000$ comparisons/s, which is trivial for one core.
 - Decision: an indexed heap or sorted set is enough at this scale
 - A timing wheel is a 10× follow-up, not a baseline need.

- History
 - Persisting every heartbeat creates 5.76 billion rows/day.
 - Persist status transitions and sampled evidence instead.
 - Keep raw heartbeat diagnostics briefly only when needed for investigation.
 - Decision: transitions are the durable record
 - Raw heartbeats get a bounded retention window measured in hours.
- Push and pull both create total work proportional to fleet size.
 - Push helps NAT traversal and distributes sending schedules.
 - It does not make central ingestion constant time for the whole fleet.
 - Decision: push is the primary channel
 - Pull is a rate-limited verifier for the suspect cohort only.

Architecture and flow — IC3 baseline



- Registry is the control plane.
 - Mechanism bought: one source of expected devices, active incarnations, and maintenance windows.
 - Cost paid: collectors must cache assignments so heartbeat processing survives a registry outage
 - Enrollment does not.
- Heartbeat ingress is the data plane entry.
 - Mechanism bought: authenticates the device credential and routes by stable device identity to one shard.
 - Cost paid: stateless fleet must be sized for the full ingest rate plus reconnect bursts after an outage.
- Shard owner serializes updates.
 - Mechanism bought: one active logical owner per shard makes version and deadline updates atomic without a distributed lock per heartbeat.
 - Cost paid: physical failover needs a new ownership epoch, and downstream must reject an old owner's epoch.
- Verifier supplies secondary evidence.
 - Mechanism bought: an independent reachable path turns some SUSPECT cases into UNREACHABLE or back to UP.
 - Cost paid: bounded concurrency and queue age
 - Same-path probes add little independent evidence.
- Transition outbox and alert grouping own downstream fan-out.
 - Mechanism bought: transitions are durable and deduplicated by (device_id, transition_version) before history and alerting consume them.
 - Cost paid: alert latency includes outbox drain
 - The outbox backlog is a monitored quantity.
- Coverage watchdog keeps monitoring self-health separate.
 - Mechanism bought: an independent check of collector coverage
 - Dashboards expose observation age.

- Cost paid: another component to keep alive
 - Last-known UP is not presented as current UP during a blind interval.
- Authenticate the device credential and resolve its registry identity.
- Validate its registered incarnation and increasing sequence.
- Set last-seen from collector receive time.
- Atomically replace its deadline and latest state.
- Emit a transition only when the observable state changes.
- When a deadline expires, compare the current version before marking suspect.
- Verify through an independent reachable path when available.
- Group alarms by shared collector, network, or region failure.

API and data model — IC3 baseline

Surface	Contract	Idempotency and conflict behavior
POST /v1/heartbeats	Device incarnation, sequence, bounded status payload	Duplicate or older sequence returns 202 and is ignored; collector supplies receive time
GET /v1/devices/{id}/health	State, observation time, monitoring coverage	Read-only; tenant scope and freshness are explicit
GET /v1/devices?state=SUSPECT	Cursor page over a tenant's current index	Read-only; bounded page size, no fleet-wide synchronous scan
PUT /v1/devices/{id}/maintenance	Suppression deadline and reason	If-Match version; mismatch returns 409; audited; observations continue
POST /v1/devices/{id}:enroll	Authorize a new device incarnation	Idempotency key; old incarnation cannot reactivate itself
POST /v1/devices/{id}:probe	Request a bounded verification	202 with operation id; duplicate request joins the pending probe

State	Key	Fields and boundary
Registry	(tenant_id, device_id)	Expected flag, region, active incarnation, maintenance, version
Latest observation	(tenant_id, device_id)	Incarnation, sequence, receive time, deadline, status, owner epoch
Deadline index	(shard_id, deadline, device_id)	Stores observation version for stale-task rejection
Transition	(device_id, transition_version)	Old/new state, cause, observation time; deduplication identity
Shard ownership	shard_id	Owner instance, epoch; only the current epoch may publish transitions

- Reference DDL
 - Keys and uniqueness match the state table above and the [companion schema](#).
 - The live latest observation and the deadline index live in shard-owner memory; SQL holds the registry, incarnations, transitions, a bounded-rate observation checkpoint, and shard ownership.

```

CREATE TABLE devices (
  tenant_id text NOT NULL, device_id text NOT NULL,
  expected boolean NOT NULL, -- false: silence is not an incident
  region text NOT NULL,
  active_incarnation text, -- NULL until first enrollment
  maintenance_until timestampz, maintenance_reason text, -- suppression only; observations continue
  credential_ref text NOT NULL, -- mTLS identity reference, never key
  material
  version bigint NOT NULL CHECK (version > 0), -- If-Match validator
  PRIMARY KEY (tenant_id, device_id),
  CHECK ((maintenance_until IS NULL) = (maintenance_reason IS NULL))
);
CREATE INDEX devices_by_region ON devices (tenant_id, region, device_id);
CREATE TABLE device_incarnations (
  tenant_id text NOT NULL, device_id text NOT NULL, incarnation text NOT NULL,
  idempotency_key text NOT NULL, enrolled_at timestampz NOT NULL,
  revoked_at timestampz, -- a revoked incarnation cannot reactivate
  itself
  PRIMARY KEY (tenant_id, device_id, incarnation),
  UNIQUE (tenant_id, device_id, idempotency_key), -- a repeat enroll returns the first result
  FOREIGN KEY (tenant_id, device_id) REFERENCES devices
);
CREATE TABLE device_transitions (
  tenant_id text NOT NULL, device_id text NOT NULL, transition_version bigint NOT NULL,
  old_state text NOT NULL CHECK (old_state IN ('UP','SUSPECT','UNREACHABLE','UNKNOWN')),
  new_state text NOT NULL CHECK (new_state IN ('UP','SUSPECT','UNREACHABLE','UNKNOWN')),
  cause text NOT NULL, observed_at timestampz NOT NULL, -- collector receive time, never device
  sentAt
  owner_epoch bigint NOT NULL CHECK (owner_epoch > 0), -- only the current epoch may publish
  PRIMARY KEY (tenant_id, device_id, transition_version), -- deduplication identity
  FOREIGN KEY (tenant_id, device_id) REFERENCES devices,
  CHECK (old_state <> new_state)
);
CREATE INDEX device_transitions_history ON device_transitions (tenant_id, device_id, observed_at,
transition_version);
CREATE TABLE device_observation_checkpoints (
  tenant_id text NOT NULL, device_id text NOT NULL,
  incarnation text NOT NULL, sequence bigint NOT NULL CHECK (sequence >= 0),
  received_at timestampz NOT NULL, deadline_at timestampz NOT NULL, -- wall clock; the live deadline is
  monotonic in memory
  state text NOT NULL CHECK (state IN ('UP','SUSPECT','UNREACHABLE','UNKNOWN')),
  observation_version bigint NOT NULL CHECK (observation_version > 0), -- stale timers compare against this
  owner_epoch bigint NOT NULL CHECK (owner_epoch > 0),
  PRIMARY KEY (tenant_id, device_id),
  FOREIGN KEY (tenant_id, device_id) REFERENCES devices,
  CHECK (deadline_at > received_at)
);
CREATE INDEX device_state_index ON device_observation_checkpoints (tenant_id, state, device_id); -- GET
/devices?state=
CREATE TABLE shard_owners (
  shard_id integer NOT NULL, owner_instance text NOT NULL,
  epoch bigint NOT NULL CHECK (epoch > 0), -- fence; publish only if epoch = expected
  lease_until timestampz NOT NULL,
  PRIMARY KEY (shard_id)
);

```

- Not captured by the DDL
 - The (shard_id, deadline, device_id) index is an in-memory ordered structure per shard owner; it is rebuilt from device_observation_checkpoints with a warm-up UNKNOWN interval and never queried per tick.
 - device_observation_checkpoints is written at a bounded checkpoint rate, not per heartbeat; the 66,667 heartbeats/s hot path stays in memory.
 - Raw heartbeats are retained briefly in a log keyed by (tenant, device); only transitions reach SQL.
- Deviation from the companion

- Companion `devices.state` moves to `device_observation_checkpoints.state` so the registry row is desired configuration and the projection is observed evidence with its own version and epoch.
- `device_incarnations` and `shard_owners` are added for the enrollment idempotency key and the ownership epoch the deep dive depends on; `device_transitions` is the companion's `device_health_events` under this file's name.
- `shard_owners` is not tenant-scoped because a shard owns devices of many tenants.
- Incarnation handling
 - A reboot may reset sequence numbers.
 - Assign or register a new incarnation through an authenticated enrollment flow.
 - Do not compare random boot UUIDs as if they were ordered epochs.
 - Reject delayed heartbeats from a revoked prior incarnation.
- Conventions
 - Receive time is the collector's clock, never the device's `sentAt`.
 - Every read carries observation age and coverage status so a stale UP is never silent.
 - Long-running probes return an operation id and are polled, not held open.

Deadline algorithm and ownership — IC3 baseline

```
HEARTBEAT(device, incarnation, seq), under shard serialization:
  require current owner epoch before publishing any observation
  reject if incarnation is not the registry's active incarnation
  ignore if seq <= latest accepted seq for that incarnation
  version += 1
  last_seen = collector_receive_time
  deadline = last_seen + timeout
  replace device's deadline-index entry
  atomically persist observation, deadline identity, and any transition outbox

EXPIRE(device, observed_version):
  under the same serialization / conditional update:
    require current owner epoch
    if current.version != observed_version: ignore stale timer
    if current.deadline > now: ignore early timer
    if collector coverage unhealthy: mark observation UNKNOWN
    otherwise persist SUSPECT and verification outbox in one conditional transaction
```

- Ordered index baseline
 - A heap with an indexed update or a sorted set supports $O(\log N)$ renewal.
 - Process only deadlines due at the current time.
 - Keep one live deadline entry per device where practical.
- Timing-wheel follow-up
 - Round deadlines upward to the next tick, never earlier than the actual deadline.
 - Check the exact deadline again when a bucket fires.
 - A 5 s tick adds at most about 5 s quantization delay under regular scheduling.
 - Rescheduling must remove old entries or compact obsolete versions.
 - Scanning all bucket keys on every tick is not a constant-time wheel.
- Collector restart
 - Restore latest observations from replicated state or a checkpoint.
 - Rebuild deadlines with an explicit warm-up/unknown interval.
 - Shared storage alone does not eliminate restart races.
 - Ownership fencing and freshness checks remain necessary.
- Clock boundary
 - Use owner-local monotonic elapsed time for live deadline intervals.
 - Store wall-clock receive time separately for display and recovery metadata.

- Monotonic timestamps cannot be compared across process restarts or hosts.
- After failover, reconstruct deadlines conservatively with a stated wall-clock skew allowance.
- If that allowance cannot be established, expose `UNKNOWN` until fresh heartbeats restore coverage.
- Exact failure if the version check is skipped
 - A timer scheduled at version 7 fires after a heartbeat advanced the device to version 8.
 - Without the compare, the device is marked `SUSPECT` 45 s after a heartbeat that arrived seconds ago.
 - With an old owner still running, the same write can also arrive from a stale epoch
 - The epoch check rejects it.

Verification and alerting — IC3 baseline

- Pull verification
 - Use only a permitted reachable path.
 - NAT may require an existing outbound control channel or a local gateway.
 - Reusing the same failed network path adds little independent evidence.
- Verification budget
 - If 1% of the fleet is suspect at once, that is 10,000 probes.
 - At an assumed verifier concurrency of 500 and 1 s per probe, the cohort drains in about 20 s.
 - A regional outage can make 10% suspect
 - The verifier must shed load and let grouping speak instead.
- Alerting
 - Suppress duplicate device pages when monitoring coverage itself fails.
 - Preserve the affected-device list for diagnosis.
 - Maintenance suppresses paging, not health recording.
 - A monitoring uncertainty signal must not trigger destructive automated failover by itself.
- Grouping rule
 - Correlate transitions within a short window by collector shard, site, and region.
 - Above a threshold share of a group, emit one incident with a device count instead of per-device pages.
 - Keep per-device transitions in history so the incident can be expanded.

Failure, operations, and security

Failure	Blast radius	Detection	Degraded behavior	Recovery
Single heartbeat lost	One device	Next sequence arrival	No immediate page	Refresh deadline on a newer heartbeat
Device unreachable	Device or access network	Silence and independent observation	<code>SUSPECT</code> or <code>UNREACHABLE</code> , not proven dead	Recover on fresh authenticated evidence
Collector shard unavailable	Its assigned fleet	Independent coverage watchdog	Mark observation <code>UNKNOWN</code> ; group monitoring alarm	Restore ownership and rebuild deadlines
Region network partition	Regional fleet	Correlated missing reports	One regional incident plus affected-device count	Reconcile latest state without replaying stale alarms
Old timer races with heartbeat	One device	Version comparison	Ignore stale expiry	Keep the new deadline
Device reboot resets sequence	One device	New enrollment/incarnation	Ignore obsolete incarnation messages	Register fresh incarnation securely
Verifier queue overloaded	Suspect cohort	Queue age	Delay secondary evidence with visible age	Rate-limit verification and prioritize critical devices

Failure	Blast radius	Detection	Degraded behavior	Recovery
Registry or metadata store unavailable	Enrollment, maintenance, assignment changes	Control API errors	Collectors keep processing heartbeats on cached assignments (static stability)	Restore store; reconcile assignment version
Transition outbox backlog	Alert and history latency	Outbox age	Dashboards show state with a lag flag	Drain; deduplicate by <code>transition_version</code>

- Observe
 - Heartbeat age percentiles.
 - Registered versus currently observed devices.
 - Collector coverage by shard.
 - Expiry work backlog.
 - Verification latency and independent success rate.
 - Correlated state transitions by network and region.
 - Outbox age and transition duplicates rejected.
- Retry discipline
 - Agents retry with exponential backoff and jitter
 - A failed heartbeat is not retried past the next scheduled one.
 - After an outage, reconnects spread over a window so ingress does not see the whole fleet at once.
 - Ingress returns `429` with a retry hint when a shard is over budget
 - Agents honor it.
 - Verifier probes retry at most once
 - A second failure is evidence, not a retry trigger.
- Tenant isolation
 - Device identity and every read are scoped by tenant
 - Another tenant's device id returns `404`, not `403`.
 - Per-tenant heartbeat rate quotas stop one tenant's misconfigured fleet from starving a shard.
 - Alert groups never span tenants even when the underlying network incident does.
 - Device credentials are per incarnation and revoked at enrollment of the next one.
- Rollout and rollback safety
 - Update a small agent cohort.
 - Jitter both steady heartbeats and reconnect backoff.
 - Compare gaps and payload rejection before fleet rollout.
 - Keep the old payload version readable during migration.
 - Collector deploys hand off shard ownership with a new epoch and a declared warm-up interval
 - Rollback follows the same handoff.

Alternatives and IC4 stretch

Option	What it is	Strength	Weakness	Choose when
Push heartbeat	Device-initiated report	NAT-friendly and naturally distributed scheduling	Agent may be alive while application is unhealthy	Primary liveness channel
Central pull	Scheduled remote probe	Observer controls checks	Reachability and probe storms	Small or directly reachable fleets
Push plus selective verification	Heartbeat with secondary observation	Better evidence on suspicious silence	Same-path verification remains correlated	Baseline when a second path exists
Fixed timeout	Deadline from last observation	Easy to explain and operate	False suspicions under variable delay	Stable heartbeat cadence
Adaptive suspicion	Threshold from latency distribution	Handles variable timing	More tuning and less predictable thresholds	Measured jitter makes fixed thresholds noisy

- IC4 stretch — partition by operational cell
 - Each cell owns collectors, hot state, and alarm grouping.
 - Global dashboards read aggregated cell summaries.
 - One overloaded cell must not delay every region's deadline processing.
- IC4 stretch — separate detection from remediation
 - Remediation checks authoritative ownership and fencing before replacing a worker.
 - A false positive must not create two active owners of external state.
- IC4 stretch — adapt detection targets by device class
 - A 60 s target cannot coexist with devices sleeping offline for hours without an explicit expected schedule.
 - Battery and bandwidth constraints may reverse the always-on-agent premise.
- IC4 stretch — 10× load
 - At 10 million devices the same interval produces approximately 666,667 heartbeats/s.
 - Expand cells and deadline ownership.
 - Move the deadline index to a timing wheel so renewal is constant time per shard.
 - Keep global alert aggregation bounded by incidents rather than raw device count.
- IC4 stretch — cost and blast radius
 - Collector cost is proportional to fleet size
 - History retention is the dominant storage cost, so raw heartbeat retention is a cost control.
 - A cell failure strands one cell's fleet in UNKNOWN, not the whole region.
 - Reconnect storms after an outage are the largest transient cost
 - Jittered reconnect is a capacity control, not a nicety.
- Weakest assumption
 - Heartbeat delivery latency stays well under the 45 s timeout, so silence reflects the device rather than network jitter.
 - Mitigation: measure interarrival distribution per device class and site
 - Move noisy classes to adaptive thresholds before lowering the fixed timeout.
- Over-engineering warning
 - Do not claim a failure detector can distinguish a crash from an arbitrary partition with certainty.
 - Do not build a timing wheel or per-device leases before the heap and version compare are shown to fall short.

Level signal — IC3 baseline vs IC4 stretch

Phase	IC3 must land	IC4 adds	Separating signal
Requirements	Registry-defined expectations, NAT constraint, 45 s suspect target	Names remediation and metrics as non-goals; device-class schedules	Separates suspicion from certainty before drawing anything
Estimates	Ingest rate, hot state, history row count	Deadline-work arithmetic that keeps the heap and rejects per-heartbeat DB writes	A number decides the hot-path storage
Architecture	Shards by device id, one owner, watchdog separate from collectors	Control plane versus data plane; cached assignments survive registry outage	States static stability without being asked
Deep dive	Version-checked expiry, incarnation plus sequence ordering	Owner epochs, monotonic versus wall-clock boundary, conservative rebuild after failover	Reasons about the same race across restarts and hosts
Operations	Failure table, coverage watchdog, grouped alerts	Reconnect-storm control, outbox lag, cell blast radius	Volunteers monitoring-of-the-monitor
Wrap	Recap and push-versus-pull tradeoff	Names delivery-latency assumption and adaptive thresholds first	Identifies the flaw before the interviewer does

Interview pushback and validation

- Q1: Does push reduce total fleet work to $O(1)$?
 - **IC3 baseline**
 - Each device does constant work, but the service receives $O(N)$ reports.
 - **IC4 stretch**
 - Hierarchical summaries reduce global traffic while regional collection remains proportional to fleet size.
- Q2: A heartbeat arrives while its timer expires.
 - **IC3 baseline**
 - Compare observation version and deadline under one atomic boundary.
 - **IC4 stretch**
 - Reject transitions from a prior shard owner as well.
- Q3: Can two failed probes confirm a dead machine?
 - **IC3 baseline**
 - They confirm failed observations, not the physical cause.
 - **IC4 stretch**
 - Require independent evidence and a fenced remediation protocol.
- Q4: Devices cannot run an agent.
 - **IC3 baseline**
 - Use pull where reachable or an on-site observer.
 - **IC4 stretch**
 - Renegotiate detection coverage and freshness where neither path exists.
- Q5: What happens when collectors restart?
 - **IC3 baseline**
 - Rebuild state and expose a warm-up unknown interval.
 - **IC4 stretch**
 - Move ownership with an epoch and reconcile transition history before paging.
- Q6: Why not persist every heartbeat?
 - **IC3 baseline**
 - The daily row count is billions
 - Transitions meet the history requirement.
 - **IC4 stretch**
 - Retain sampled raw evidence only for a bounded forensic window.
- Q7 (premise reversal): Why not write `last_seen` to the database and scan for stale rows?
 - **IC3 baseline**
 - That is 66,667 row writes/s plus a 1 million-row scan every tick
 - The database becomes the hot path and the scan bounds detection latency.
 - **IC4 stretch**
 - The database is a checkpoint for recovery
 - The deadline index is in-memory per shard owner and fenced by epoch.
- Q8: Can the device's `sentAt` drive the deadline?
 - **IC3 baseline**
 - No
 - Device clocks are unordered across the fleet, so collector receive time drives timeouts.
 - **IC4 stretch**
 - Keep `sentAt` only to measure agent-side delay and detect clock drift as a diagnostic.
- Validation scenarios
 - Expiry racing with refresh.
 - Timer tick rounding.

- Old incarnation replay after reboot.
- Clock regression on collector host.
- Collector outage causing mass silence.
- Verifier unavailable behind NAT.
- Ownership transfer with delayed old-owner writes.
- Registry outage during steady-state heartbeats.
- Reconnect storm after a regional partition heals.

Recall card

- "The registry defines expected devices; heartbeats establish recent observations."
- "A versioned deadline prevents stale expiry from overwriting fresh evidence."
- "Silence produces suspicion, while monitoring outages produce visible uncertainty."
- "Independent verification and grouped alerts improve diagnosis without pretending to prove a crash."
- "Collectors keep serving on cached assignments when the registry is down; that is the static-stability line."
- Wrap-up script
 - "The design is a registry of expected devices, heartbeats sharded by device id to one owner, an in-memory deadline index, and a separate coverage watchdog."
 - "The key invariant is that expiry is version-checked and epoch-fenced, so a stale timer or an old owner never overwrites fresh evidence."
 - "The main tradeoff is push for NAT-friendliness and distributed scheduling versus pull for observer control; I use push with bounded verification."
 - "My weakest assumption is heartbeat delivery latency well under the timeout; I would measure interarrival jitter per device class before trusting the 45 s threshold."

References

- [Ranked question and report provenance](#).
- [OCI role framing](#).
- [Detailed API and schema companion](#).
- Reference check: 2026-09-09.
 - Official documentation supports the mechanisms cited next to each link.
 - Capacity values and architecture choices remain proposed interview assumptions.
- [OCI load-balancer health configuration](#): an example of configured health observations.
- [Envoy outlier detection](#): local-origin versus externally observed failures.
- The heartbeat state machine and timing estimates are derived design choices, not documented OCI internals.